

When Post-Hoc Calibration Hurts: Platt Scaling and Isotonic Regression on Modern Tabular Models

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Abstract

Post-hoc calibration is routinely applied on the assumption that it cannot degrade a classifier. We test that over 3,150 runs per calibrator: 21 classifiers, 30 binary TabArena-v0.1 tasks, 5-fold cross-validation. Under log-loss, two of the most widely used methods are unreliable: isotonic regression has a mean relative change of +3.22% and improves log-loss in only 43.7% of runs, and Platt scaling improves it in 49.8%, showing no consistent directional advantage. Venn–Abers attains the largest mean reduction (−14.17%) and Beta calibration the highest improvement rate (67.1%). Mean AUC-ROC changes by under 1% throughout and mean accuracy is essentially unaffected except under Pearsonify, so changes in probabilistic quality are largely invisible in discrimination and classification metrics.

Keywords: Probability calibration, Venn–Abers predictors, conformal prediction, proper scoring rules, tabular data, benchmarking.

1. Motivation

Calibrated probabilities underpin reliable decision making in risk-sensitive domains (Chen et al., 2018; Fonseca and Lopes, 2017). Platt scaling (Platt, 1999) and isotonic regression (Zadrozny and Elkan, 2002) have been the standard remedies since Niculescu-Mizil and Caruana (2005), and are applied largely without verification. Venn–Abers predictors (Vovk et al., 2014, 2015) instead inherit distribution-free validity from conformal prediction. Since that evidence predates modern tabular architectures, we re-examine whether post-hoc calibration reliably improves proper scoring performance.

2. Experimental design

We benchmark 21 classifiers — classical baselines, tree ensembles (CatBoost, XGBoost, LightGBM, EBM) and neural or foundation models for tabular data (ModernNCA, TabICL, TabPFN v2, TabM) — on the 30 binary tasks of TabArena-v0.1 (Erickson et al., 2025) under 1×5 -fold stratified cross-validation, giving 3,150 runs per calibrator. The five model-agnostic calibrators, fitted on a stratified split held out from the training folds, are isotonic regression, Platt scaling, Beta calibration in its three-parameter **abm** form (Kull et al., 2017), Venn–Abers, and Pearsonify (xRiskLab, 2025), a recent method based on Pearson residuals. Per evaluation we compute the relative change $\Delta = 100 (M_{\text{cal}} - M_{\text{base}}) / M_{\text{base}}$ against the uncalibrated model, summarised by its mean $\hat{\mathbb{E}}[\Delta]$ and improvement rate; $\Delta < 0$

is an improvement for losses. Primary measures are log-loss and Brier score, strictly proper rules (Gneiting and Raftery, 2007), with AUC-ROC, accuracy and compute alongside and thresholds at 0.5.

3. Results

Table 1: Mean relative change $\hat{\mathbb{E}}[\Delta]$ and improvement rate $\hat{\mathbb{P}}(\Delta < 0)$ over all 3,150 runs per calibrator.

Calibrator	Log-loss		Brier score		AUC-ROC
	$\hat{\mathbb{E}}[\Delta]$	improved	$\hat{\mathbb{E}}[\Delta]$	improved	$\hat{\mathbb{E}}[\Delta]$
Isotonic regression	+3.22%	43.7%	−3.74%	52.5%	−0.43%
Platt scaling	−9.75%	49.8%	−1.88%	43.0%	−0.05%
Beta calibration	−13.70%	67.1%	−3.91%	57.5%	+0.06%
Venn–Abers	−14.17%	63.2%	−4.14%	54.2%	−0.09%
Pearsonify	+22.90%	20.7%	+30.69%	10.9%	−0.92%

Two standard methods are unreliable under log-loss. Isotonic regression has a positive mean relative log-loss change and improves log-loss in fewer than half of all runs. Platt scaling improves it in 49.8%, showing no consistent directional advantage despite a favourable mean driven by a heavy left tail. Isotonic’s mean changes have opposite signs under log-loss and Brier score, while Platt improves both means but not most evaluations. This reflects their weighting of errors: Brier is bounded and quadratic, log-loss heavily penalises near-degenerate incorrect probabilities.

The effect is strongly model-dependent. Aggregate means conceal substantial inter-model variance: for LDA, Beta and Venn–Abers reduce mean log-loss by roughly 10% while isotonic and Pearsonify raise it by 14.8% and 27.4%. Among the strongest uncalibrated models — TabICL, CatBoost and EBM, all in the top four by log-loss rank — the uncalibrated model often outranks all five calibrated counterparts. Venn–Abers attains the largest mean improvement on both proper scores and is the only method with no relative log-loss degradation exceeding 50%; Beta has the highest improvement rate on both. AUC-ROC declines in 65–76% of runs. Inference cost rises in 73.0% of runs under Venn–Abers but in only about half under each of the other four (Manokhin and Grønhaug, 2026).

4. Discussion

Post-hoc calibration is not cost-free: substantial changes in probabilistic quality can be largely invisible to discrimination and classification metrics, so any calibrator should be validated on the base model against a strictly proper scoring rule. The calibration split reduces the data available to the base learner, so these results compare complete pipelines under a fixed data budget rather than calibrators applied to an unchanged model — the choice a practitioner faces, but one that leaves the calibrator’s effect inseparable from the smaller training set.

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